
Pediatric erythromelalgia: A retrospective review of 32 cases evaluated at Mayo Clinic over a 37-year period

Robert H. Cook-Norris, MD,^a Megha M. Tollefson, MD,^a Andres E. Cruz-Inigo, BA,^c Paola Sandroni, MD, PhD,^b Mark D. P. Davis, MD,^a and Dawn M. R. Davis, MD^a
Rochester, Minnesota

Background: Erythromelalgia has not been well characterized in the pediatric population.

Objective: We sought to review our experience of erythromelalgia in the pediatric age group.

Methods: We conducted a retrospective review of patients 18 years of age and younger with a diagnosis of erythromelalgia who were examined at Mayo Clinic in Rochester, MN, from 1970 to 2007.

Results: The records of 32 patients (girls, 22 [69%]) were evaluated. Mean age was 14.1 years (range, 5-18 years) and mean time to diagnosis was 5.2 years. Seven patients (22%) had a first-degree relative with erythromelalgia; 4 were from the same family. Physical activity was limited because of discomfort in 21 patients (66%) and school attendance was affected in 11 patients (34%). Noninvasive vascular studies, which compared temperature, laser Doppler flow, and transcutaneous oximetry in the toes, identified vascular abnormalities in 13 (93%) of 14 patients. Neurophysiologic studies with autonomic reflex screening (including quantitative sudomotor axon reflex test and thermoregulatory sweat testing) showed evidence of a small-fiber neuropathy involving the skin in 10 (59%) of 17 patients studied; there was no evidence of large-fiber neuropathy in 20 patients in whom electromyographic and nerve conduction studies were performed. Topical lidocaine was the most commonly prescribed treatment (44%). Fifteen patients were monitored for an average of 9.1 years (median, 5.0 years; range, 0.4-23.7 years). At last follow-up, 5 patients had stable disease, 4 showed improvement, two had resolution, one reported worsening of symptoms, and 3 had died (one suicide).

Limitations: Conclusions are limited because this was a retrospective chart review.

Conclusion: Erythromelalgia in pediatric patients is associated with substantial morbidity and even death. The majority of cases are not inherited. Most patients studied have associated small-fiber neuropathy. The disease course is variable. A reliable and safe treatment has not been determined. (J Am Acad Dermatol 2012;66:416-23.)

Key words: acrocyanosis; cellulitis; erythromelalgia; erythromelalgia; neuropathy; neurophysiology; pediatric psychology; Raynaud phenomenon; small-fiber neuropathy; vascular anomalies.

From the Department of Dermatology,^a Department of Neurology,^b and College of Medicine,^c Mayo Clinic.

The first two authors contributed equally to this article.

Funding sources: None.

Conflicts of interest: None declared.

Accepted for publication July 27, 2011.

Reprint requests: Dawn M. R. Davis, MD, Department of Dermatology, Mayo Clinic, 200 First St SW, Rochester, MN 55905. E-mail: davis.dawnmarie@mayo.edu.

Published online July 28, 2011.

0190-9622/\$36.00

© 2011 by the American Academy of Dermatology, Inc.

doi:10.1016/j.jaad.2011.01.010

Erythromelalgia is a clinical syndrome distinguished by the occurrence of red, hot extremities (Fig 1). The term was coined by Mitchell¹ in 1878 and is derived from 3 Greek words—"erythros" (red), "melos" (extremity), and "algos" (pain). Primary (idiopathic) and secondary (associated with an underlying disease) forms have been described. Although fairly well studied in adults, the demographic features, natural history, prognosis, characteristics, and pathogenesis of erythromelalgia are poorly characterized in the pediatric age group. There is increasing recognition that most adult

patients have electrophysiologic evidence of an associated small-fiber neuropathy, but no studies have reported this finding in children.

Experience of pediatric patients with idiopathic, noninherited erythromelalgia has been documented primarily in anecdotal case reports and in a few larger series of both pediatric and adult cases.²⁻⁴ Nine pediatric patients with secondary erythromelalgia and associated hypertension were studied by Drenth et al⁵ in 1995; 8 of these cases were previously published as case reports. Inherited primary erythromelalgia is well described in families with an autosomal dominant mode of inheritance.⁶⁻⁹ Mutations in *SCN9A*, which encodes the sodium channel protein Na(v)1.7 subunit, have been identified in cases of inherited erythromelalgia, whereas these mutations have rarely been reported in noninherited disease.¹⁰

We undertook a retrospective chart review of pediatric patients with erythromelalgia seen at our institution to examine the demographic characteristics, presentation, functional impairment, frequency and type of vascular and neurologic abnormalities, and response to therapy.

METHODS

The Mayo Clinic Institutional Review Board approved this study. It involved all pediatric patients (age 0 months-18 years) with a master index diagnosis of erythromelalgia or erythralgia as determined by a dermatologist, rheumatologist, geneticist, or vascular medicine physician at Mayo Clinic in Rochester, MN, from January 1970 through May 2007. Patients who denied research authorization were excluded. We reviewed 50 charts, and only patients who met the diagnostic criteria were included in the study. Six of these patients were reported in a previously published case series.³ As proposed by Mitchell,¹ our definition of erythromelalgia was the occurrence of erythema, heat (observed subjectively, objectively, or both), and associated discomfort (variously described as pain, burning, or tingling) in the extremities. These signs and symptoms were noted either during the physical examination or in the medical history, because of the intermittent nature of the disease.

Demographics, clinical presentation, and comorbidities

Abstracted data included sex, age at onset of symptoms, time to diagnosis, and earlier diagnoses and treatments. Charts were reviewed for symptom characteristics and distribution, associated illnesses, family history of erythromelalgia, functional impairment, physical findings, and laboratory tests.

Functional assessment

Functional assessment included evaluation of physical activity and school performance. These topics are addressed on the standardized forms completed for all pediatric patients presenting to our institution. Guardians are requested to answer questions regarding the amount of time spent on physical activity, difficulty performing age-appropriate activities, school absence caused by illness,

and whether the patient's current grade level is appropriate for his or her age.

Neurophysiologic and vascular testing

Vascular testing was performed both in the presence and absence of symptoms and the results were compared as previously described.^{11,12} Measurements of toe temperature, laser Doppler flow, and transcutaneous oximetry were performed.

Neurophysiologic testing included evaluation of both large-fiber nerves (electromyographic and nerve conduction studies) and small-fiber nerves (autonomic reflex screen, including quantitative sudomotor axon reflex test^{11,12} and thermoregulatory sweat testing as previously described).¹³

Serologic testing: the paraneoplastic autoantibody comprehensive panel

Tests for antineuronal nuclear antibodies type 1 (sometimes called *anti-Hu*) and type 2 (or *anti-Ri*), amphiphysin antibody, Purkinje cell cytoplasmic antibody (PCA) type 1 (sometimes called *anti-Yo*), PCA type 2, PCA-Tr, and collapsin response-mediator protein antibody were done as a one-step indirect immunofluorescence screening assay.¹⁴⁻¹⁶ PCA type 1 is associated with breast and ovarian carcinomas, PCA-Tr with Hodgkin lymphoma, antineuronal nuclear antibodies type 2 with breast and small cell lung carcinomas, and all other autoantibodies with small cell lung carcinoma. Tests for antineuronal calcium channel

CAPSULE SUMMARY

- Erythromelalgia is characterized by red, hot, painful extremities.
- The majority of pediatric cases are not inherited.
- Neurophysiologic studies showed evidence of small-fiber neuropathy in 59% of patients studied.
- Marked morbidity and increased mortality were observed.
- Lidocaine patches appear to be a potentially useful and safe therapy.



Fig 1. Red, hot, and painful feet of patient with erythromelalgia.

antibodies of P/Q-type and N-type specificities were immunoprecipitation assays.¹⁷ These assays constitute the paraneoplastic autoantibody comprehensive panel performed by Mayo Medical Laboratories in Rochester, MN, and are a component of the paraneoplastic autoantibody lung panel.¹⁸

Treatment and follow-up

We recorded all interventions implemented after presentation to our institution and the response to treatment as documented in follow-up visits or correspondence. Response to treatment was documented as marked improvement, some improvement, or no improvement in symptoms on the basis of history or physical examination findings. Patients were defined as lost to follow-up when there was no recorded follow-up either as a clinic visit or by correspondence at least 1 month after implementation of therapy.

RESULTS

Patient demographics

Thirty-two pediatric cases were identified in the 37-year study period. Patient demographics are summarized in Table I. Seven patients, of whom 4 were from the same family, had a first-degree relative with erythromelalgia. Fifteen patients (47%) had a diagnosis at presentation. The most common earlier diagnosis was erythromelalgia (6 patients [19%]), followed by neuropathy, cellulitis, and Raynaud phenomenon (two patients [6%] each). Other diagnoses included reflex sympathetic dystrophy, adverse medication effect, unspecified vasospastic disease, and foot pain.

Treatment types and reported response at initial presentation are summarized in Table II. Twenty-three (72%) patients had received one or more therapies. Aspirin and nonsteroidal anti-inflammatory medications had been used most commonly (each in 44% of patients), followed by antidepressants (31%). Antihistamines, vasodilators, β -blockers, narcotic analgesics, gabapentin, and corticosteroids had been prescribed less frequently.

Table I. Demographics of 32 patients with erythromelalgia

Characteristics	Value
Sex, No. (%)	
Female	22 (69)
Male	10 (31)
Age at diagnosis, y	
Mean	13.8
Median	15.4
Range	0.3-17.9
Time to diagnosis, y	
Mean	4.6
Median	3.1
Range	0.2-13.9
Patients with family history of erythromelalgia (n = 7; 22%)	
Age at diagnosis, y	
Mean	6.2
Median	7.3
Range	0.3-16.1
Time to diagnosis, y	
Mean	6.5
Median	6.8
Range	0.7-11.0

Clinical presentation and comorbidities

Clinical presentation and physical findings are summarized in Table III. Erythromelalgia symptoms were intermittent in 26 patients (81%). Although the feet were involved in all 32 patients, the legs were also affected in 15 (47%) and the hands in 15 (47%). Other involved areas included the face (5 patients), ears (two patients), and nose (one patient). Numbness was not reported by any patient. Affected extremities were often described as discolored with a purple hue, cool, and mildly painful. Twenty-four patients (75%) had clinical findings consistent with erythromelalgia; physical examination was unremarkable in 8 (25%) (Table III). No patient had a history of myeloproliferative disease, and none had clinical features suggestive of a syndromic disorder.

Functional impact

Physical activity was limited in most patients (66%), with 4 patients (13%) being wheelchair-bound (Table IV). School attendance was adversely affected in 11 children (34%). Depression, anxiety, and behavior problems were encountered in 9 patients (28%).

Diagnostic investigations

Two (7%) of 28 patients had an abnormal complete blood cell count, with thrombocytosis in one patient and anemia in the other. Erythrocyte sedimentation

Table II. Response to earlier medications and other treatments in 23 patients presenting to Mayo Clinic with erythromelalgia

Therapeutic intervention	Response, No. of patients			
	n	Very helpful	Somewhat helpful	Not helpful
Aspirin	14	1	2	11
Nonsteroidal anti-inflammatory drugs (ibuprofen, indomethacin, naproxen)	14	1	0	13
Antidepressants (amitriptyline hydrochloride, nortriptyline hydrochloride, fluphenazine hydrochloride, cyproheptadine hydrochloride, venlafaxine hydrochloride)	10	2	2	6
Antihistamines (diphenhydramine hydrochloride, cetirizine hydrochloride, cimetidine)	8	0	0	8
Vasodilators (nitroprusside sodium, nifedipine, diltiazem hydrochloride)	7	1	1	5
β -Blockers (propranolol hydrochloride, atenolol, nadolol)	6	0	1	5
Narcotics (codeine phosphate, morphine sulfate, fentanyl citrate)	6	0	4	2
Gabapentin	6	2	0	4
Parenteral corticosteroids (oral, intramuscular, intravenous)	6	0	1	5
Topical corticosteroids	5	0	0	5
Physical methods (biofeedback, intrathecal pump, TENS unit)	4	1	1	2
Other anticonvulsants (carbamazepine, phenytoin)	3	0	1	2
Sympathectomy	3	0	1	2
Acetaminophen	3	1	0	2
Doxazosin mesylate	1	0	0	1
Capsaicin	1	0	0	1
Ergotamine tartrate	1	0	0	1
Mexiletine hydrochloride	1	0	0	1
Clonidine hydrochloride	1	0	1	0
Tetracycline	1	0	0	1
Homeopathic (pulsatilla, belladonna)	1	0	0	1

TENS, Transcutaneous electrical nerve stimulation.

rate was within normal limits in all 22 patients for whom it was checked. Serum protein electrophoresis was normal in 14 patients. Antinuclear antibody levels were elevated in 6 of 13 patients who had documented connective tissue disease serology (antinuclear antibodies and extranuclear antibodies). Paraneoplastic serology showed negative results for all 6 patients tested. Skin biopsy specimens were obtained in 8 patients; in 7 patients, histopathologic findings showed nonspecific inflammation, and in one patient, there was evidence of leukocytoclastic vasculitis. Genetic testing for mutations in the sodium channel Na(v)1.7 was not performed.

Neurophysiologic and vascular testing

The results of vascular and neurophysiologic studies are reported in Table V. Vascular tests performed when symptoms were present showed a markedly increased temperature, noted in 12 (92%)

of 13 symptomatic patients, compared with when symptoms were absent. Two (67%) of 3 symptomatic patients who had laser Doppler flow recorded had increased blood flow. Transcutaneous oximetry indicated decreased transcutaneous pressure of oxygen in 4 (50%) of 8 symptomatic patients.

Neurophysiologic testing showed no evidence of large-fiber neuropathy in the 12 patients who had electromyographic and nerve conduction studies. Seventeen patients underwent tests of small-fiber nerve function. The majority of the results were abnormal: 4 of 6 (67%) had abnormal quantitative sudomotor axon response test results and 6 of 14 (43%) had abnormal thermoregulatory sweat testing.

Treatment

Twenty-seven patients (84%) received one or more forms of medical therapy at our institution. Fourteen (43%) were treated with topical lidocaine

Table III. Clinical presentation and physical examination findings

Findings	Patients, No. (%) (N = 32)
Clinical presentation	
Symptom duration	
Intermittent	26 (81)
Constant	6 (19)
Location of involvement	
Feet	32 (100)
Legs	15 (47)
Hands	15 (47)
Face	5 (16)
Ears	2 (6)
Patient symptoms	4 (13)
Red	32 (100)
Painful	31 (97)
Hot	28 (88)
Burning	19 (59)
Swollen	13 (41)
Numbness	0 (0)
Symptom exacerbation	
Exercise	26 (81)
Heat	21 (66)
Time of day (night)	15 (47)
Standing	14 (44)
Symptom relief	
Cooling	23 (72)
Leg elevation	8 (25)
Physical examination	
Suggestive of erythromelalgia	24 (75)
Redness	21 (66)
Warmth	14 (44)
Edema	10 (31)
Coolness of affected limb	6 (19)
Acrocyanosis	4 (13)
Ulcers	5 (16)
Reticular cutaneous pattern	7 (22)
Unremarkable	8 (25)
Comorbidities	
Hypertension	2 (6)
Anemia	2 (6)
Obesity	1 (3)
Diabetes type 1	1 (3)
Myeloproliferative disorder	0 (0)

patches. One of these cases was previously published.¹⁹ Five reported marked improvement, one reported no improvement, and 8 were lost to follow-up. Eight patients (25%) were treated with gabapentin; 3 reported no change in symptoms and the other 5 were lost to follow-up. Eight patients (25%) were treated with aspirin: two reported marked improvement; one, some improvement; and one, no improvement. The other 4 patients were lost to follow-up. Five patients (16%) received a topical gel formulation of 1% amitriptyline and 0.5%

Table IV. Functional assessment of study patients

Function	Patients, No. (%) (N = 32)
Physical activity	
Not affected/no limitations	11 (34)
Moderately affected/physical activity limited	13 (40)
Severely affected/intolerant of physical activity	4 (13)
Wheelchair-bound	4 (13)
Schooling	
Regular attendance	21 (66)
Numerous absences	4 (13)
Not attending school	3 (9)
Homeschooled because of condition	4 (13)
Mental health	
Not affected	23 (72)
Affected	9 (28)
Depression only	3 (9)
Anxiety only	1 (3)
Behavior problems only	2 (6)
Anxiety and depression	2 (6)
Depression and behavior problems	1 (3)

ketamine, one of whom was previously described²⁰; one reported marked benefit, one reported no change in symptoms, and 3 were lost to follow-up. Of 4 patients (13%) treated with tricyclic antidepressants (amitriptyline or nortriptyline), two had marked improvement, one no change, and one was lost to follow-up. Three patients were treated with anticonvulsant medications (carbamazepine, phenytoin, or topiramate), two with no improvement, and one (treated with carbamazepine) with a marked improvement. For two patients, misoprostol therapy resulted in no change in symptoms. Phenoxybenzamine, fluphenazine, and epidural block were administered to one patient each, with a marked improvement recorded. Diazepam, propranolol, plasma exchange, and biofeedback were administered to one patient each, with some reported improvement. Cyclophosphamide, dapsone, mexiletine, topical lidocaine gel, and sympathectomy (type not specified) were administered to one patient each without benefit.

Follow-up

Of the 27 patients who received treatment at our institution, 12 (44%) were lost to follow-up. The other 15 patients were monitored for an average of 9.1 years (median, 5.0 years; range, 0.4-23.7 years). At last follow-up, 5 patients had stable disease, 4 had experienced improvement, two had resolution of disease, one reported worsening of symptoms, and 3

Table V. Vascular and neurophysiologic testing of study patients

	No. tested with symptoms/total tested (%)	No. positive/total tested with symptoms (%)	No. abnormal/total tested (%)
Vascular studies during symptoms vs without symptoms			
Increased temperature	13/20 (65)	12/13 (92)	NA
Increased laser Doppler flow	3/5 (60)	2/3 (67)	NA
Decreased TcPo ₂ value	8/15 (53)	4/8 (50)	NA
Small-nerve fiber testing			
Thermoregulatory sweat testing	NA	NA	6/14 (43)*
QSART	NA	NA	4/6 (67)*
Large-nerve fiber testing			
EMG/nerve conduction studies	NA	NA	0/12 (0)

EMG, Electromyogram; NA, not applicable; QSART, quantitative sudomotor axon reflex test; TcPo₂, transcutaneous pressure of oxygen.

*17 Patients/20 tests.

had died. Deaths were a result of suicide (n = 1), sepsis (n = 1), and bone-marrow suppression secondary to cyclophosphamide therapy for vasculitis (n = 1). An affected first-degree relative (sister) of each of two unrelated patients had committed suicide, reportedly because of erythromelalgia.

DISCUSSION

Erythromelalgia is a rare syndrome. In the only population-based study yet published, the age- and sex-adjusted incidence rate was determined to be 1.3 per 100,000 people per year.¹⁴ In the pediatric subgroup, the incidence was lower still, and there were no cases of inherited erythromelalgia in the cohort reported.

Consistent with reports in adults,²⁻⁴ the condition was more common in females; the female to male ratio was 2.2:1. The syndrome often goes unrecognized, as reflected by the mean time to diagnosis of 5 years in the current series. The clinical presentation was consistent with that described in adult patients: during symptomatic periods, the affected areas were red, hot, and painful³; between these episodes, they were frequently cool, cyanotic, and mildly uncomfortable.¹⁵

Underlying disorders previously described in association with erythromelalgia^{2,3,5,16,17} were not present in this group of pediatric patients; in particular, none had myeloproliferative disease. Only one patient had hypertension. In a previous report, 7 of 9 pediatric patients with erythromelalgia had underlying hypertension.⁵

The increased morbidity associated with pediatric erythromelalgia is apparent in this patient cohort. Eleven patients (34%) had difficulty with truancy, including prolonged leaves of absence from school or cessation of attendance. Some patients made the transition to a homeschool curriculum. Patients who

attended school regularly often reported altering their activities to alleviate or prevent symptoms. Removing socks or shoes, limiting physical activity, or avoiding participation in physical education or sports (despite having a desire to do so) was a common occurrence. Four patients had such debilitating disease that they required a wheelchair. Several patients also had depression, anxiety, fatigue, or trouble sleeping.

Virtually all patients went to extreme measures to improve their symptoms, with such commonly described activities as going barefoot, keeping the house as cool as possible, using fans, cooling affected areas with cold water and ice, resting, and elevating the legs. Some reported soaking their feet in cold water for prolonged periods; occasionally, patients described doing this for more than 20 hours a day (even while sleeping). This practice often resulted in immersion injury leading to maceration, ulceration, and infection.

Our study found that erythromelalgia is associated not only with increased morbidity but also with death. Three patients (9%) died, with one death each caused by suicide, sepsis (the patient died, at age 15 years, of sepsis as a consequence of prolonged soaking of the extremities), and bone-marrow failure induced by cyclophosphamide treatment for vasculitis. In addition, one affected first-degree relative of each of two unrelated patients committed suicide, reportedly because of erythromelalgia.

The marked morbidity and increased mortality have been previously noted in adults, with suicide a notable cause of death in adults also.³

Measures of vascular function during episodes of erythromelalgia were compared with measurements taken between symptomatic periods and showed the increased temperature and increased blood flow previously observed in adults.^{11,12} Neurophysiologic

assessment in the majority of patients demonstrated decreased sweating indicative of a small-fiber neuropathy on quantitative sudomotor axon reflex test and thermoregulatory sweat testing, as previously described in adults.¹¹⁻¹³ The large-fiber neuropathy determined by electromyographic and nerve conduction studies in adult patients was not found.

In common with previous reports,²¹ histopathologic examination of skin biopsy specimens did not show distinctive findings in the majority (7 of 8 patients).

Treatment possibilities

No treatment was a panacea for any patient. Although various medications had been administered before presentation (Table II), most were at best partially helpful. Subsequently prescribed medications were of variable benefit. This variability in the response to medications has been well documented in previous publications from studies of adults.^{3,22} We noted that lidocaine patches were helpful for pain relief in many patients, as has been reported in adult patients at our institution.²³ Other agents and treatment methods described in case reports of erythromelalgia—such as intravenous lidocaine and oral mexiletine in combination, sodium nitroprusside, recombinant growth hormone, regional anesthesia blockade, and thalamic stimulation^{20,24-29}—were not used in our patients.

Study strengths

The report has several strengths. It involved a large cohort of patients with this extremely rare disorder, and we present aspects of the illness that have not been described previously in children.

Study limitations

A retrospective study of a rare, poorly defined clinical syndrome at a tertiary referral center has multiple inherent limitations. Referral bias and survival bias are likely, leading to the inclusion of potentially more severe or refractory cases and of patients seen late in the course of their disease. Because some patients travel a great distance for evaluation at our institution, follow-up can be challenging, as evidenced by nearly half of our treated patients being lost to follow-up.

Genetic studies were not performed because testing was not clinically available at the time the patients were evaluated. Studies have shown that mutations in *SCN9A*, which encodes the sodium channel protein Na(v)1.7 subunit,⁷ are present in some patients with inherited erythromelalgia but are rare in noninherited cases of erythromelalgia.¹⁰ Molecular genetic testing for mutations in *SCN9A* is

now available and can be considered for patients with inherited disease. Patients with a *SCN9A* mutation may have onset of symptoms in adolescence.⁸ Alternative splicing could explain why members of a family with inherited erythromelalgia may have symptom onset at a variable age.

CONCLUSION

In summary, we present a large cohort of patients in the pediatric age group with erythromelalgia diagnosed at Mayo Clinic. The disease went unrecognized for a mean of 5 years. Erythromelalgia had a marked negative impact on daily activities and was associated with increased mortality and suicide in one patient. Small-fiber neuropathy was associated with the disorder. No single medication was successful in controlling the disease.

REFERENCES

- Mitchell SW. On a rare vaso-motor neurosis of the extremities, and on the maladies with which it may be confounded. *Am J M Sc* 1878;76:17-36.
- Kurzrock R, Cohen PR. Erythromelalgia: review of clinical characteristics and pathophysiology. *Am J Med* 1991;91:416-22.
- Davis MD, O'Fallon WM, Rogers RS III, Rooke TW. Natural history of erythromelalgia: presentation and outcome in 168 patients. *Arch Dermatol* 2000;136:330-6.
- Kalgaard OM, Seem E, Kvernebo K. Erythromelalgia: a clinical study of 87 cases. *J Intern Med* 1997;242:191-7.
- Drenth JP, Michiels JJ, Ozsoylu S. Erythromelalgia Multidisciplinary Study Group. Acute secondary erythromelalgia and hypertension in children. *Eur J Pediatr* 1995;154:882-5.
- Drenth JP, Finley WH, Breedveld GJ, Testers L, Michiels JJ, Guillet G, et al. The primary erythromelalgia-susceptibility gene is located on chromosome 2q31-32. *Am J Hum Genet* 2001;68:1277-82.
- Choi JS, Dib-Hajj SD, Waxman SG. Inherited erythromelalgia: limb pain from an S4 charge-neutral Na channelopathy. *Neurology* 2006;67:1563-7.
- Cheng X, Dib-Hajj SD, Tyrrell L, Waxman SG. Mutation I136V alters electrophysiological properties of the Na(v)1.7 channel in a family with onset of erythromelalgia in the second decade. *Mol Pain* 2008;4:1.
- Choi JS, Cheng X, Foster E, Leffler A, Tyrrell L, Te Morsche RH, et al. Alternative splicing may contribute to time-dependent manifestation of inherited erythromelalgia. *Brain* 2010;133:1823-35.
- Han C, Rush AM, Dib-Hajj SD, Li S, Xu Z, Wang Y, et al. Sporadic onset of erythromelalgia: a gain-of-function mutation in Nav1.7. *Ann Neurol* 2006;59:553-8.
- Davis MD, Sandroni P, Rooke TW, Low PA. Erythromelalgia: vasculopathy, neuropathy, or both? A prospective study of vascular and neurophysiologic studies in erythromelalgia. *Arch Dermatol* 2003;139:1337-43.
- Sandroni P, Davis MD, Harper CM, Rogers RS III, Harper CM Jr, Rogers RS III, et al. Neurophysiologic and vascular studies in erythromelalgia: a retrospective analysis. *J Clin Neuromuscul Dis* 1999;1:57-63.
- Davis MD, Genebriera J, Sandroni P, Fealey RD. Thermoregulatory sweat testing in patients with erythromelalgia. *Arch Dermatol* 2006;142:1583-6.

14. Reed KB, Davis MD. Incidence of erythromelalgia: a population-based study in Olmsted County, Minnesota. *J Eur Acad Dermatol Venereol* 2009;23:13-5.
15. Davis MD, Wilkins F, Rooke TW. Between episodes of erythromelalgia: a spectrum of colors. *Arch Dermatol* 2006;142:1085-6.
16. Kurzrock R, Cohen PR. Erythromelalgia and myeloproliferative disorders. *Arch Intern Med* 1989;149:105-9.
17. Babb RR, Alarcon-Segovia D, Fairbairn JF II. Erythromelalgia: review of 51 cases. *Circulation* 1964;29:136-41.
18. Lee HR, Lennon VA, Camilleri M, Prather CM. Paraneoplastic gastrointestinal motor dysfunction: clinical and laboratory characteristics. *Am J Gastroenterol* 2001;96:373-9.
19. Davis MD, Sandroni P. Lidocaine patch for pain of erythromelalgia. *Arch Dermatol* 2002;138:17-9.
20. Sandroni P, Davis MD. Combination gel of 1% amitriptyline and 0.5% ketamine to treat refractory erythromelalgia pain: a new treatment option? *Arch Dermatol* 2006;142:283-6.
21. Davis MD, Weenig RH, Genebriera J, Wendelschafer-Crabb G, Kennedy WR, Sandroni P. Histopathologic findings in primary erythromelalgia are nonspecific: special studies show a decrease in small nerve fiber density. *J Am Acad Dermatol* 2006;55:519-22.
22. Kurzrock R, Cohen PR. Classification and diagnosis of erythromelalgia. *Int J Dermatol* 1995;34:146-7.
23. Davis MD, Sandroni P. Lidocaine patch for pain of erythromelalgia: follow-up of 34 patients. *Arch Dermatol* 2005;141:1320-1.
24. Cimaz R, Rusconi R, Fossali E, Careddu P. Unexpected healing of cutaneous ulcers in a short child. *Lancet* 2001;358:211-2.
25. Delye H, Lagae L, Vermeylen J, Nuttin B. Thalamic stimulation as a treatment for primary erythromelalgia: technical case report. *Neurosurgery* 2005;57(Suppl):E404.
26. Harrison CM, Goddard JM, Rittey CD. The use of regional anesthetic blockade in a child with recurrent erythromelalgia. *Arch Dis Child* 2003;88:65-6.
27. McGraw T, Kosek P. Erythromelalgia pain managed with gabapentin. *Anesthesiology* 1997;86:988-90.
28. Nathan A, Rose JB, Guite JW, Hehir D, Milovcich K. Primary erythromelalgia in a child responding to intravenous lidocaine and oral mexiletine treatment. *Pediatrics* 2005;115:e504-7.
29. Rauck RL, Naveira F, Speight KL, Smith BP. Refractory idiopathic erythromelalgia. *Anesth Analg* 1996;82:1097-101.